

UNIVERSITÀ DEGLI STUDI DI PADOVA

DIPARTIMENTO DI INGEGNERIA CIVILE EDILE E AMBIENTALE

Corso di Laurea in Mathematical Engineering

Classe LM-44



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



**Numerical Investigation of Vortex
Shedding Past Bluff Bodies**

Applied Computational Fluid Dynamics

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1 Introduction

The flow around bluff bodies is a classical and widely studied problem in computational fluid dynamics due to its relevance in many engineering applications, such as aerodynamics, wind engineering and structural dynamics. A characteristic feature of such flows is the formation of alternating vortices in the wake, commonly referred to as vortex shedding. Vortex shedding induces unsteady aerodynamic forces on the body, which may lead to flow-induced vibrations, increased drag and fatigue issues in engineering structures. The shedding process is typically characterized by the Strouhal number, which relates the dominant shedding frequency to the characteristic length scale of the body and the incoming flow velocity.

Despite the simplicity of the governing equations, the flow past bluff bodies exhibits complex unsteady behavior that strongly depends on the geometry of the obstacle. Even small variations in shape can significantly alter the wake structure, the vortex shedding frequency and the resulting aerodynamic forces. For this reason, simplified two-dimensional configurations are often employed to investigate the fundamental mechanisms governing wake dynamics.

In this work, a numerical investigation of vortex shedding behind three different bluff body geometries is performed: a circular cylinder, a square cylinder and a hexagonal cylinder. All simulations are carried out at the same Reynolds number and under identical flow conditions in order to isolate the effect of the body geometry on the flow dynamics.

The simulations are performed using the open-source multiphysics framework *Kratos Multiphysics*, adopting a monolithic finite element formulation with Variational Multiscale (VMS) stabilization. The unsteady flow behavior is analyzed through time histories of the lift and drag coefficients, frequency analysis of the lift signal and visualization of the wake structures.

The remainder of this report is organized as follows. Section 2 presents the numerical methodology, including governing equations, discretization and boundary conditions. Section 3 discusses the numerical results, focusing on the vortex shedding characteristics and aerodynamic coef-

ficients for the different geometries. Finally, conclusions are drawn in Section 4.

2 Problem Description

The problem consists of an incompressible viscous flow past a 2D bluff body placed inside a channel. Three geometries are considered:

- Circular cylinder
- Square cylinder
- Hexagonal cylinder

The characteristic length D is defined as the diameter of the circular cylinder ($D = 200mm$) and as the equivalent width for the other geometries. The flow is characterized by a Reynolds number defined as:

$$Re = \frac{UD}{\nu}, \quad (1)$$

where U is the reference inlet velocity (inlet mean velocity equal to $1m/s$ and ν is the kinematic viscosity of the fluid ($\nu = 0.002 \text{ m}^2/s$).

In the present study, the Reynolds number is fixed at $Re = 100$, corresponding to an unsteady laminar regime characterized by periodic vortex shedding. This allows a direct comparison of the wake dynamics and aerodynamic coefficients induced solely by geometric effects.

3 Numerical Methodology

3.1 Governing equations

The flow is assumed to be incompressible and Newtonian. The governing equations are the incompressible Navier–Stokes equations, which express the conservation of mass and momentum:

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} \quad (3)$$

where $\mathbf{u}$ is the velocity vector, p the pressure, ρ the fluid density and ν the kinematic viscosity.

3.2 Spatial&Time discretization

The governing equations are discretized in space using the finite element method. A monolithic velocity–pressure formulation is adopted, in which both variables are solved simultaneously.

Stabilization is achieved through the Variational Multiscale (VMS) approach. Both the standard VMS and the quasi-static VMS (QSVMS) formulations are employed for comparison. The comparison between VMS and QSVMS formulations is performed in order to assess the sensitivity of the results to the stabilization strategy. These stabilized formulations allow the use of same order of interpolation for velocity and pressure while preventing numerical instabilities associated with convection-dominated flows.

Time integration is performed using an implicit scheme. The time step Δt is selected in order to properly ensure a sufficient temporal resolution of the shedding frequency, corresponding to several time steps per shedding period. and numerical stability.

The simulations are run for a sufficiently long physical time to allow the flow to reach a fully developed periodic regime. Initial transient effects are discarded in the post-processing phase.

4 Computational Setup

4.1 Computational mesh

The computational domain consists of a two-dimensional channel containing a bluff body. Three different geometries are considered: a circular cylinder, a square cylinder and a hexagonal cylinder.

An unstructured mesh is employed, with refinement in the vicinity of the obstacle and in the wake region in order to accurately capture the vortex shedding dynamics. The minimum element size near the body is chosen based on the characteristic dimension of the geometry, obtaining in total, around 10000 elements. This mesh resolution represents a compromise between computational cost and the ability to resolve the unsteady wake dynamics.

4.2 Boundary and initial conditions

At the inlet, a prescribed velocity profile is imposed. A parabolic inlet velocity profile is imposed and smoothly ramped in time during the initial transient, to avoid numerical instabilities.

No-slip boundary conditions are enforced on the surface of the obstacle. At the outlet, a do-nothing boundary condition is applied, allowing the flow to exit the domain without artificial constraints.

The initial condition corresponds to a quiescent flow field.

4.3 Fluid Properties

The fluid is assumed to be Newtonian and incompressible. The density and dynamic viscosity are chosen such that the desired Reynolds number is obtained.

4.4 Time-Stepping Parameters

The time step Δt is selected to satisfy stability and accuracy requirements. Simulations are run for a sufficiently long physical time to allow the flow to reach a statistically periodic regime before collecting data.

5 Post-Processing and Data Analysis

The aerodynamic forces acting on the obstacle are monitored during the simulation. The drag and lift coefficients are defined as:

$$C_D = \frac{F_x}{\frac{1}{2}\rho U^2 D}, \quad C_L = \frac{F_y}{\frac{1}{2}\rho U^2 D} \quad (4)$$

where F_x and F_y are the drag and lift forces, U is the reference inlet velocity and D is the characteristic dimension of the body.

The vortex shedding frequency is obtained from the lift coefficient signal using spectral analysis and peak detection. The Strouhal number is then computed as:

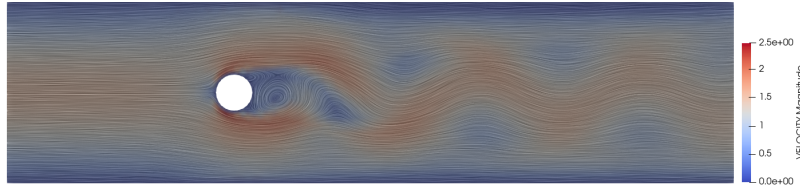
$$St = \frac{fD}{U} \quad (5)$$

The root-mean-square (RMS) value of the lift coefficient is also evaluated to quantify the intensity of the unsteady flow oscillations.

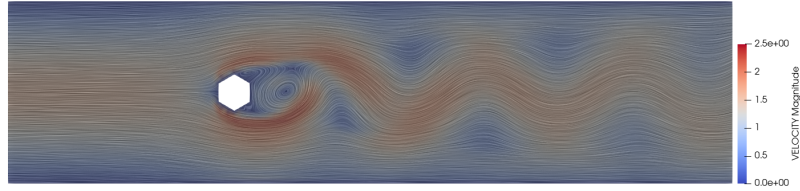
6 Results and Discussion

6.1 Flow Field and Vortex Shedding

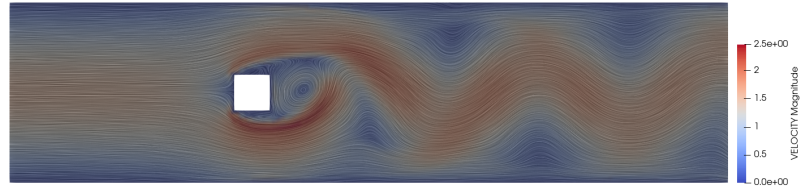
The instantaneous velocity and vorticity fields reveal the formation of a Von Kármán vortex street downstream of all bluff bodies. The circular cylinder exhibits a well-organized and symmetric vortex street, while the square and hexagonal geometries generate a more irregular wake, characterized by stronger vortices and a wider recirculation region.



(a) Circular cylinder



(b) Hexagonal cylinder



(c) Square cylinder

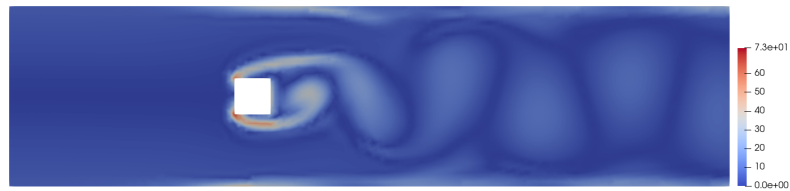
Figure 1: Instantaneous velocity fields for the three bluff body geometries at the same phase of the shedding cycle.



(a) Circular cylinder



(b) Hexagonal cylinder



(c) Square cylinder

Figure 2: Instantaneous vorticity fields for the three bluff body geometries at the same phase of the shedding cycle.

6.2 Aerodynamic Coefficients

Figure 3 shows the time history of the lift coefficient for the three geometries. The circular cylinder exhibits a nearly sinusoidal signal, while sharper geometries produce higher lift fluctuations due to stronger flow separation and more intense vortex shedding, as confirmed by the higher RMS values of the lift coefficient.

Table 1 summarizes the main aerodynamic quantities. The circular cylinder shows the lowest mean drag coefficient and lift fluctuations, while the square cylinder presents the highest drag and strongest unsteady forces. The hexagonal geometry exhibits an intermediate behavior between the

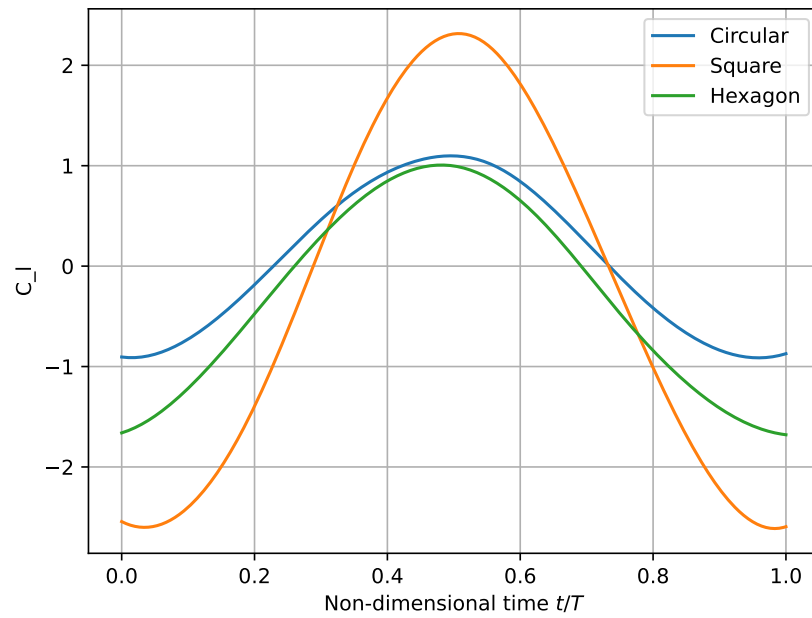


Figure 3: Lift coefficient during one period for the three geometries.

two.

Geometry	St	C_D (mean)	C_L (RMS)
Circular	0.222	2.119	0.713
Square	0.182	4.294	1.737
Hexagon	0.250	3.099	0.946

Table 1: Summary of aerodynamic coefficients and Strouhal number.

7 Conclusions and Future Work

In this work, unsteady CFD simulations of the flow past three bluff body geometries were performed using Kratos Multiphysics. The influence of the obstacle shape on vortex shedding frequency and aerodynamic forces was investigated at a fixed Reynolds number.

The results demonstrate that the geometry of the body has a significant impact on the wake dynamics. Sharper geometries lead to stronger lift fluctuations, higher drag coefficients and modified Strouhal numbers when compared to the circular cylinder. The obtained Strouhal numbers are in agreement with values commonly reported in the literature for bluff body flows at comparable Reynolds numbers. Overall, this study highlights the importance of geometric effects in bluff body aerodynamics and demonstrates the capability of Kratos Multiphysics to accurately capture vortex shedding phenomena.

Future work may include mesh refinement studies to assess numerical convergence, as well as three-dimensional simulations to capture span-wise instabilities. Additionally, the use of advanced turbulence modeling approaches such as URANS or LES could provide further insight into higher Reynolds number regimes.